

CSE250: Circuits and Electronics

Spring 2023

Practice Problems Set 3

Thevenin's Theorem, Norton's Theorem, and Maximum Power Transfer Theorem

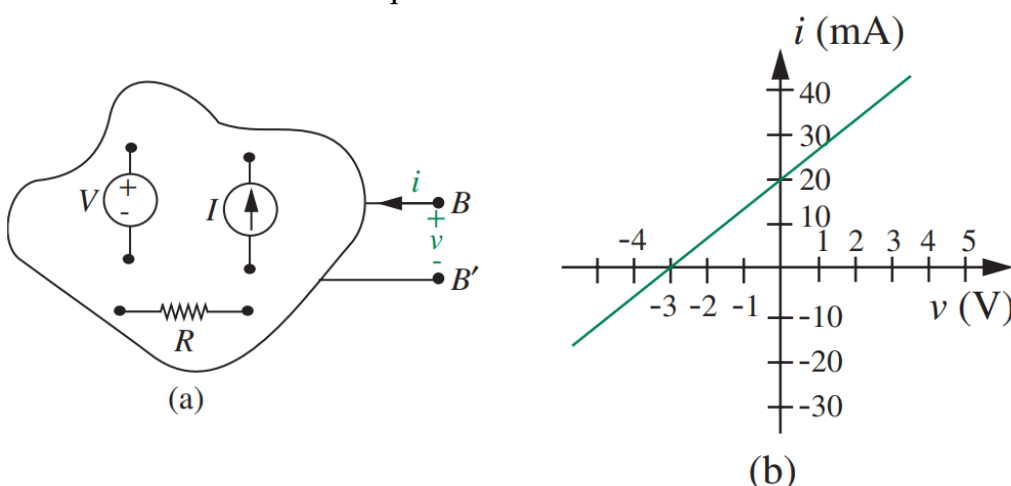
1. Measurements made on terminals $B - B'$ of a linear circuit in (a), which is known to be made up only of independent voltage sources and current sources, and resistors, yield the current-voltage characteristics shown in figure (b).

Answer:

$$V_{Th} = -3 \text{ V}$$

$$R_{Th} = \frac{3}{20} \text{ k}\Omega$$

Find the Thevenin equivalent of this circuit.

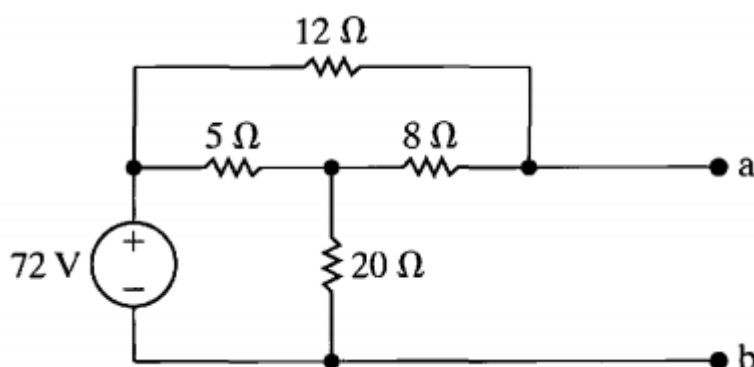


2. Find the Thevenin equivalent circuit with respect to the terminals a, b for the circuit shown.

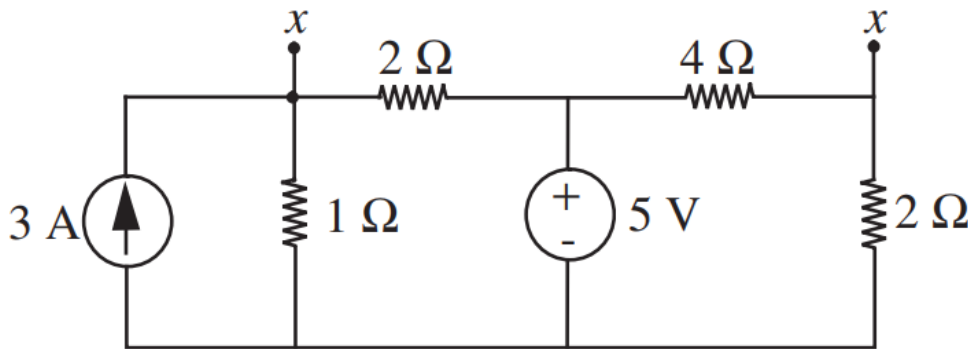
Answer:

$$V_{Th} = 64.8 \text{ V}$$

$$R_{Th} = 6 \Omega$$



3. Find the Norton equivalent at the terminals marked $x - x$ in the circuit below.

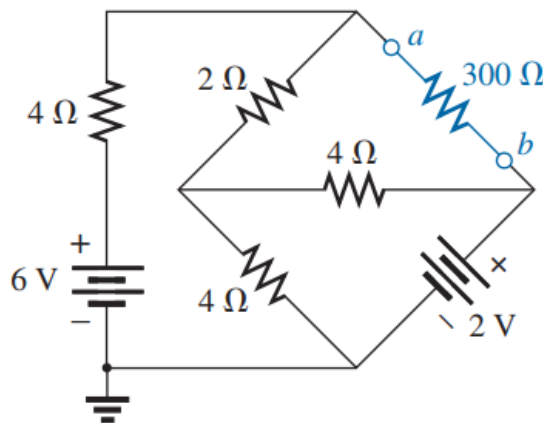


Answer:

$$V_{Th} = 2 \text{ V}$$

$$R_{Th} = 2 \Omega$$

4. For the network shown below, find the Thevenin equivalent circuit for the network external to the 300Ω resistor.

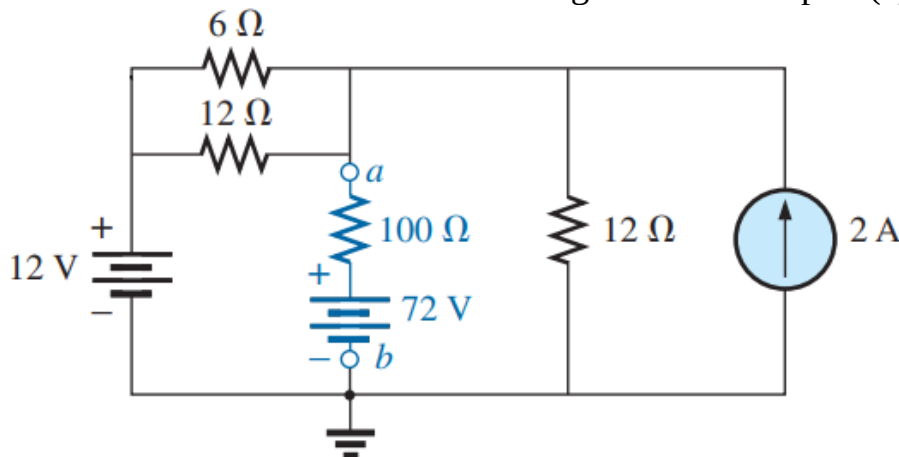


Answer:

$$V_{Th} = -6 \text{ V}$$

$$R_{Th} = 2 \Omega$$

- 5.
- Find the Norton equivalent circuit external to points a and b .
 - Find the magnitude and polarity of the voltage across the 100Ω resistor using the results of part (a).



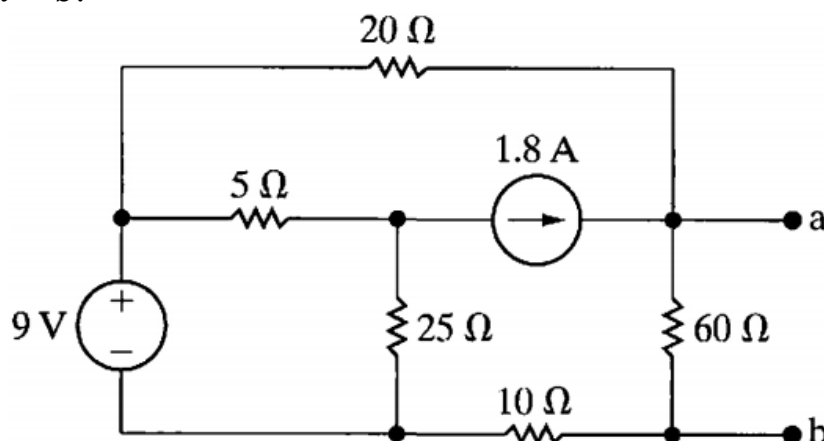
Answer:

$$I_N = 5 \text{ A}$$

$$R_{Th} = 3 \Omega$$

$$V_{100\Omega} = \mp 55.34 \text{ V}$$

6. Find the Thevenin equivalent with respect to the terminals $a - b$.

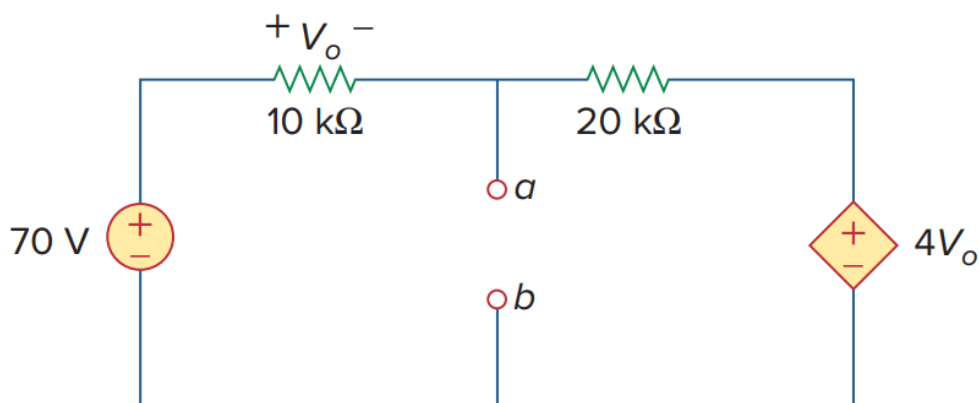


Answer:

$$V_{Th} = 30 \text{ V}$$

$$R_{Th} = 20 \Omega$$

7. Find the Thevenin equivalent at terminals $a - b$ of the circuit shown below.

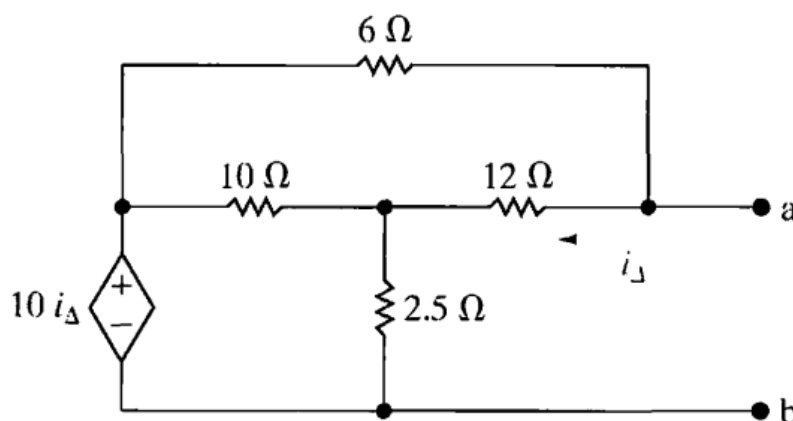


Answer:

$$V_{Th} = 60 \text{ V}$$

$$R_{Th} = \frac{20}{7} \text{ k}\Omega$$

8. Find the Thevenin equivalent with respect to the terminals $a - b$.



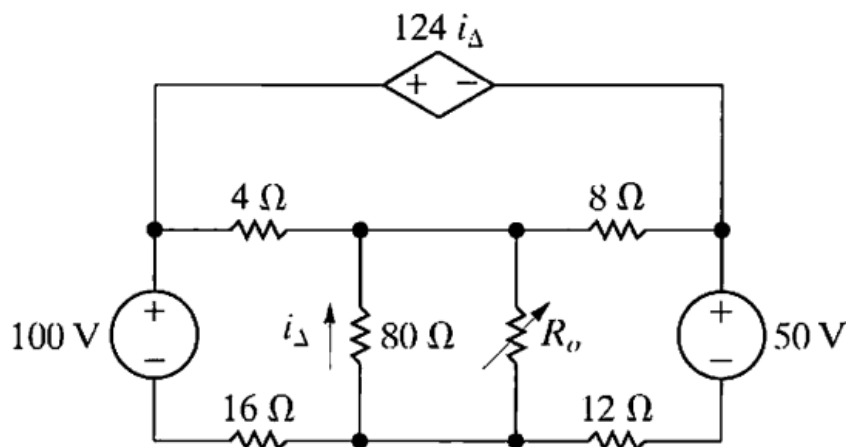
Answer:

$$V_{Th} = 0 \text{ V}$$

$$R_{Th} = 8 \Omega$$

9. The variable resistor (R_o) is adjusted for maximum power transfer to R_o .

- Find the value of R_o .
- Find the maximum power that can be delivered to R_o .

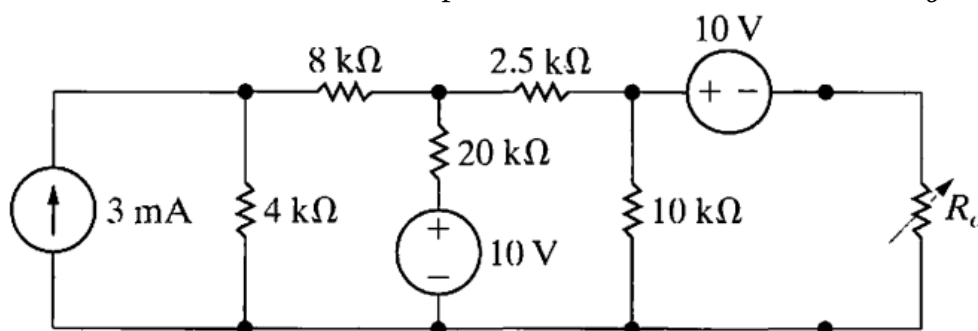


Answer:

- $R_o = 6.4 \text{ k}\Omega$
- $P_{max} = 0.90 \text{ W}$

10. The variable resistor (R_o) is adjusted for maximum power transfer to R_o .

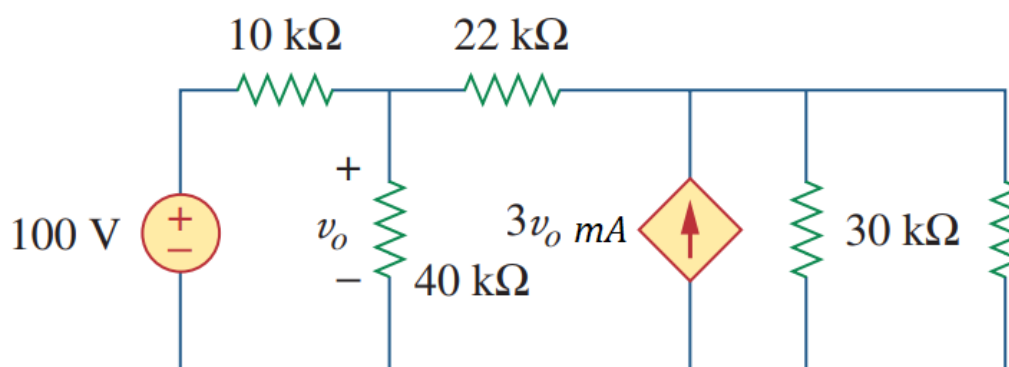
- Find the value of R_o .
- Find the maximum power that can be delivered to R_o .



Answer:

- $R_o = 5 \text{ k}\Omega$
- $P_{max} = 0.957 \text{ mW}$

11. Find the maximum power transferred to resistor R_L .



Answer:

- $$R_L = -0.65 \text{ k}\Omega$$
- $$P_{max} = \infty$$
- (Theoretically)